

Disentangling Representation Evolution in Transformers through Directional Decomposition

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Abstract

Transformer layers evolve representations through learned additive transformations that combine direction-preserving and direction-changing effects. We study this geometry by decomposing learned updates into parallel and perpendicular components. Across pre-trained models, we find substantial learned parallel components, showing that direction-preserving behavior is not confined to the residual identity path. To test whether these components are redundant or behaviorally important, we apply the decomposition to residual-space updates and to value aggregation inside attention. Inference-time edits reveal a consistent asymmetry: perpendicular edits are disruptive, while parallel edits are comparatively more redundant, especially for the self-value-aligned component of attention aggregation in value space. As supporting applications, we use the same decomposition to diagnose compression error and construct matched component-weighted pruning variants that preserve downstream quality across five models. In from-scratch pretraining, suppressing value-space parallel attention components lowers fixed-validation loss in all 12 matched architecture–seed pairs; larger single-run studies also improve average downstream scores. Overall, editing provides the central causal evidence, while compression and training test how the same geometry can inform diagnosis and intervention.

1 Introduction

Transformer layers evolve representations through learned additive updates (Vaswani et al., 2017; Elhage et al., 2021) on top of residual paths (He et al., 2016). Geometrically, each update has two

roles relative to the current representation: a parallel component that preserves direction and a perpendicular component that redirects it. This view frames representation evolution as a functional geometry problem: decomposing learned updates into these roles and testing their behavioral sensitivity under targeted edits.

Our measurements across pretrained models show that learned updates contain substantial components parallel to the current hidden state. Although a residual identity path can realize such direction preservation through a simple operation (He et al., 2016; Elhage et al., 2021), the same behavior also appears inside learned layer updates.

To test whether these components are functionally redundant or behaviorally important, we apply the decomposition in two settings. In *residual space*, we edit the updates added back to the hidden state. Given the sequence modeling inside attention, we also use a *value-space* view of value aggregation before the output projection, where the self-value contribution is separated from cross-token flow. We then use component-scaling experiments to modify parallel or perpendicular parts of the update and measure the resulting degradation.

These edits separate value-space parallel manipulation from its residual-space analogues. Scaling the part of attention aggregation aligned with the token’s own projected value keeps behavior close to baseline over a broad range. In contrast, analogous residual-space attention scaling is less stable, residual MLP-side scaling is more fragile, and perpendicular scaling is much more disruptive. Together, these comparisons show that robustness is not a generic property of being parallel: it depends on which representation space defines the component and which parts of the update are preserved.

We trace this asymmetry to value aggregation inside attention, where the self-value-aligned contri-

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bution can be targeted before projection. Residual-space analogues instead act after projection against a different reference, so similar-looking parallel-removal operations can solve different geometric objectives and produce very different benchmark behavior, including in settings related to self-directed value flow.

Finally, we test whether this geometry extends beyond inference-time edits. For compression, we decompose the error between baseline and compressed-model updates, complementing prior work on post-training compression and layer-dropping analyses of transformer redundancy (Frantar et al., 2023; Lin et al., 2024; Sun et al., 2024; Men et al., 2024; He et al., 2026). Perpendicular error tracks total compression error closely, while parallel error is less discriminative; a norm-matched isotropic control shows that much of this alignment is expected from high-dimensional concentration. We therefore treat the decomposition as a component-resolved diagnostic rather than a universally superior compression score. For training, we run from-scratch pretraining with parallel attention suppression in residual and value spaces. Across four model sizes and three matched seeds, the value-space variant lowers fixed-validation loss in all 12 architecture-seed pairs at the final tested checkpoint. Larger 1.4B and 2.7B single runs also improve average downstream scores, with the value-space variant giving the largest gain. Together, these results make editing the central causal analysis and treat compression and training as supporting applications of the same geometry.

In summary, the contributions of this work are as follows:

1. This work reveals substantial direction-preserving components in transformer updates and maps their robustness across spaces.
2. For attention updates, the value-space decomposition separates the self-value-aligned direction from the remaining value aggregation, explaining the stability gap between value-space and residual-space edits.
3. In compression and training, we test two supporting consequences of the editing result: component-resolved error diagnosis and targeted self-value-parallel suppression, with claims scoped to the controls and runs reported here.

2 Related Work

Residual-stream geometry and residual scaling.

Residual connections preserve the input representation while allowing each block to add a learned transformation (He et al., 2016). In transformers, this creates a residual stream that carries information across layers and serves as the workspace for attention and feed-forward updates (Elhage et al., 2021). This preservation path is also a practical target for architectural scaling mechanisms: ReZero (Bachlechner et al., 2021), LayerScale (Touvron et al., 2021), and deep-transformer initialization or normalization schemes (Xiong et al., 2020; Wang et al., 2024) adjust residual branch scale to improve optimization and stability. Residual paths can also mitigate degeneracy in deep self-attention, including rank collapse (Dong et al., 2021). These works establish the importance of residual pathways and residual magnitude. We study a complementary geometric property of the learned update itself: whether it preserves the current direction or rotates the representation into a perpendicular component.

Context modeling and attention geometry.

Attention explicitly aggregates information across tokens (Vaswani et al., 2017), but attention mass can also concentrate on special or persistent positions, as shown by attention-sink behavior in long-context inference (Xiao et al., 2024). Gated attention variants mitigate such sink behavior by modulating attention outputs (Qiu et al., 2025). However, even when attention sinks are reduced, large diagonal attention weights can still route substantial mass back to the current token itself. Exclusive Self-Attention (XSA) directly addresses this self-directed pathway by modifying value flow aligned with the current token (Zhai, 2026). We revisit this issue from a parallel/perpendicular geometric perspective, focusing on when attention updates preserve the current direction and when they introduce direction-changing information.

Activation steering instead adds a task- or contrast-derived vector to a residual-stream activation to control model behavior (Turner et al., 2023). Our intervention is not defined by a target behavior or contrastive prompt: it uses the model’s current hidden state or self-value as an endogenous reference and scales the analytically determined parallel and perpendicular components.

3 Background

3.1 Parallel and Perpendicular Decomposition

Let $\mathbf{z}, \Delta \in \mathbb{R}^d$ with $\mathbf{z} \neq \mathbf{0}$. The update Δ decomposes uniquely into a component parallel to $\mathbf{z}$ and a component perpendicular to it:

$$\Delta_{\parallel} = \frac{\Delta \cdot \mathbf{z}}{\|\mathbf{z}\|^2} \mathbf{z} = \alpha \mathbf{z}, \quad \alpha = \frac{\Delta \cdot \mathbf{z}}{\|\mathbf{z}\|^2}, \quad (1)$$

$$\Delta_{\perp} = \Delta - \Delta_{\parallel}, \quad \Delta_{\perp} \perp \mathbf{z}. \quad (2)$$

The parallel component $\Delta_{\parallel}$ rescales $\mathbf{z}$ without changing its direction. The perpendicular component $\Delta_{\perp}$ lives in the $(d-1)$ -dimensional subspace orthogonal to $\mathbf{z}$ and carries the direction-changing degrees of freedom.

This decomposition is the sole mathematical tool used throughout the paper. In §4 we apply it at two levels within a transformer: to the block’s residual output in the hidden-state space (§4.1), and to value aggregation in the attention’s internal representation space (§4.2).

3.2 Transformer Residual Updates

We consider transformer residual streams with L layers and hidden dimension d . Each residual sub-layer adds an update to the current representation:

$$\mathbf{z}^{l+1} = \mathbf{z}^l + \Delta^l, \quad (3)$$

The update Δ^l may denote the output of a multi-head self-attention (Attn) sub-layer, a feed-forward (FFN) sub-layer, or their combined block-level contribution. We write $\mathbf{z}_t^l$ for the hidden state at layer l , token position $t \in \{1, \dots, T\}$, and Δ_t^l for the corresponding residual update.

Using eq. 1, the parallel part of a residual update is

$$\Delta_{t,\parallel}^l = \alpha_t^l \mathbf{z}_t^l, \quad \alpha_t^l = \frac{\Delta_t^l \cdot \mathbf{z}_t^l}{\|\mathbf{z}_t^l\|^2}, \quad (4)$$

so the parallel component is equivalent to a token-wise scaling of the existing hidden state. The residual update can therefore be written as

$$\mathbf{z}_t^{l+1} = (1 + \alpha_t^l) \mathbf{z}_t^l + \Delta_{t,\perp}^l, \quad \Delta_{t,\perp}^l \perp \mathbf{z}_t^l. \quad (5)$$

To measure update geometry, we report the relative magnitude $\|\Delta_{t,\parallel}^l\|/\|\Delta_{t,\perp}^l\|$ across layers. This ratio quantifies how much of a residual update rescales the current representation (parallel) versus how much redirects it (perpendicular).

This empirical pattern, with substantial parallel components across model architectures, motivates

the functional analyses that follow: when these components affect model behavior and in which representation spaces those effects are robust.

4 Method

We instantiate the decomposition from §3.1 in two spaces: the residual stream, where it defines a block-level observable, and the attention value space, where it isolates an attention-side source of parallel updates. These two views lead to the component-scaling interventions used in the experiments. Figure 2 summarizes the resulting analysis pipeline and how it connects to editing, compression, and pretraining.

4.1 Residual-Level Decomposition

Instantiation. Set $\Delta = \Delta_t^l$ (the residual update of block f^l at token t) and $\mathbf{z} = \mathbf{z}_t^l$ (the pre-update hidden state). For the bounded intervention and localization diagnostic, we report calibration-corpus averages of the per-token fraction of update magnitude in the parallel component:

$$r_t^l = r(\Delta_t^l, \mathbf{z}_t^l) = \frac{|\alpha_t^l| \|\mathbf{z}_t^l\|}{\|\Delta_t^l\|}, \quad (6)$$

$$\alpha_t^l = \frac{\Delta_t^l \cdot \mathbf{z}_t^l}{\|\mathbf{z}_t^l\|^2}.$$

This residual-level view serves as the shared observable for the rest of the paper. Figure 1 instead displays the complementary unbounded ratio $\|\Delta_{\parallel}\|/\|\Delta_{\perp}\|$; the two quantities are not numerically interchangeable. The resulting diagnostic is block-agnostic. We apply it to (i) the Attn sub-layer output, (ii) the FFN sub-layer output, and (iii) their joint effect at the transformer-block level.

4.2 Value-Space Decomposition

Instantiation. Inside attention, each source token s is mapped to a value vector $\mathbf{v}_s = W_V \mathbf{z}_s$. For query token t , attention aggregates these value vectors before the output projection:

$$\mathbf{o}_t = \sum_{s \leq t} A_{ts} \mathbf{v}_s. \quad (7)$$

Here $\mathbf{o}_t$ denotes the pre- W_O attention output in value space. We decompose this aggregate relative to the token’s own value direction:

$$\mathbf{o}_{t,\parallel} = \frac{\mathbf{o}_t^\top \mathbf{v}_t}{\|\mathbf{v}_t\|^2} \mathbf{v}_t, \quad \mathbf{o}_{t,\perp} = \mathbf{o}_t - \mathbf{o}_{t,\parallel}. \quad (8)$$

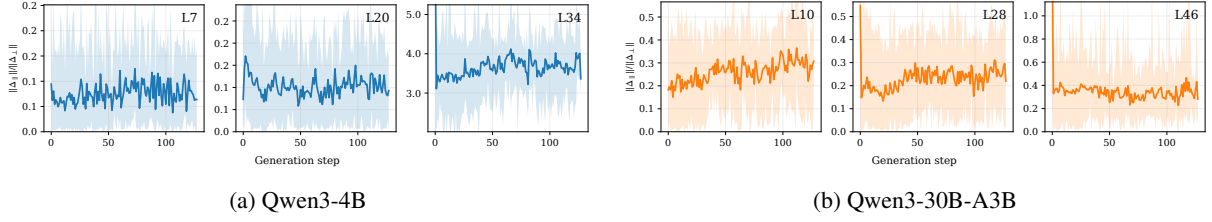


Figure 1: **Parallel components are substantial and consistent.** Across depth in both Qwen3-4B (dense) and Qwen3-30B-A3B (MoE), the ratio $\|\Delta_{t,||}^l\|/\|\Delta_{t,\perp}^l\|$ shows that parallel (rescaling) components persist alongside perpendicular (direction-changing) ones, even in deep models. This pattern motivates the functional analysis of these components in the experiments.

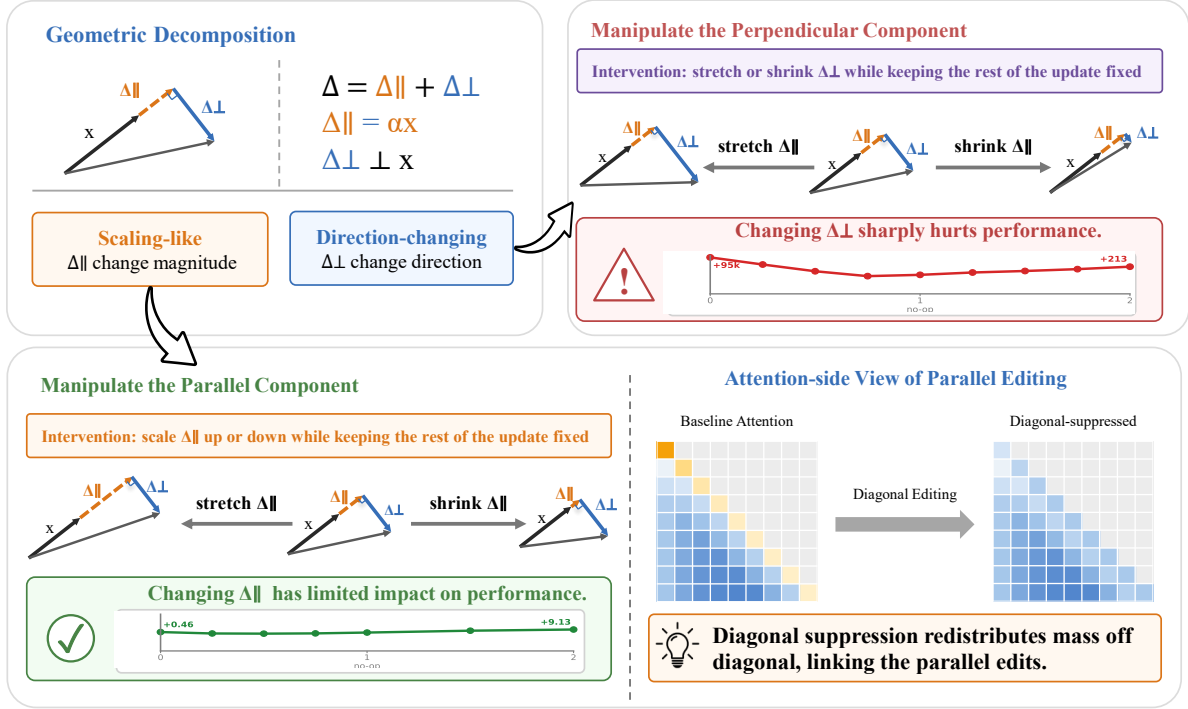


Figure 2: **Overview of the method and its downstream use.** We first measure residual-update geometry at the block level, then analyze attention aggregation in value space before output projection. This yields a unified component-scaling form for value- and residual-space edits, which we use in the experiments on inference-time editing, compression error, and pretraining behavior.

This value-space view measures how much of the pre-projection attention message follows the self-value direction versus other value-space directions. Residual-level geometry alone cannot identify this structure because the output projection has already mixed the aggregated value message into the residual stream.

Raw and exclude-self variants. Equation 8 defines the *raw* variant, which decomposes the complete value aggregate. Our primary *exclude-self* variant instead separates the direct self message

from the non-self context:

$$\begin{aligned} \mathbf{o}_t &= \underbrace{A_{tt}\mathbf{v}_t}_{\mathbf{d}_t \text{ (self)}} + \underbrace{\sum_{s < t} A_{ts}\mathbf{v}_s}_{\mathbf{c}_t \text{ (non-self)}} \\ \mathbf{c}_{t,||} &= \frac{\mathbf{c}_t^\top \mathbf{v}_t}{\|\mathbf{v}_t\|^2} \mathbf{v}_t, \quad \mathbf{c}_{t,\perp} = \mathbf{c}_t - \mathbf{c}_{t,||}. \end{aligned} \quad (9)$$

We scale only this non-self decomposition and then restore the unchanged self message:

$$\tilde{\mathbf{o}}_t^{\text{excl}} = \mathbf{d}_t + s^{(||)} \mathbf{c}_{t,||} + s^{(\perp)} \mathbf{c}_{t,\perp}. \quad (10)$$

Thus, “exclude-self” names the aggregate being decomposed: it neither masks the self-attention edge nor renormalizes the attention row. For multi-head

attention, this split and decomposition are applied independently within each head before W_O .

We apply this value-space analysis specifically to attention. The feed-forward block also produces residual-level parallel and perpendicular components, but its intermediate activation has a different channel dimension and is not an attention aggregation space. We therefore treat MLP outputs at the residual level, while using value-space decomposition for attention-side updates.

4.3 Component Scaling and Diagonal Form

For the raw value-space and residual-space interventions at query position t , we express component suppression or amplification with a general parallel-perpendicular rescaling form:

$$\begin{aligned} \mathbf{y}_t &= \sum_{s \leq t} A_{ts} \mathbf{m}_s, \\ \mathbf{y}_{t,\parallel} &= \frac{\mathbf{y}_t^\top \mathbf{r}_t}{\|\mathbf{r}_t\|^2} \mathbf{r}_t, \quad \mathbf{y}_{t,\perp} = \mathbf{y}_t - \mathbf{y}_{t,\parallel}. \end{aligned} \quad (11)$$

where $\mathbf{m}_s$ is the message vector being aggregated and $\mathbf{r}_t$ is the reference vector whose direction is used to define parallelness. The value- and residual-space interventions use different choices of these quantities (Δ_t^{Attn} denotes the post- W_O attention branch output):

$$\begin{aligned} \text{V-space:} \quad & \mathbf{m}_s = \mathbf{v}_s, \quad \mathbf{r}_t = \mathbf{v}_t, \\ & \mathbf{y}_t = \mathbf{o}_t, \quad \tilde{\mathbf{y}}_t = \tilde{\mathbf{o}}_t, \\ \text{Res. space:} \quad & \mathbf{m}_s = W_O \mathbf{v}_s, \quad \mathbf{r}_t = \mathbf{z}_t, \\ & \mathbf{y}_t = \Delta_t^{\text{Attn}}, \quad \tilde{\mathbf{y}}_t = \tilde{\Delta}_t^{\text{Attn}}. \end{aligned} \quad (12)$$

For multi-head attention, the value-space quantities are defined separately for each head before W_O . In the residual-space equations, $\mathbf{v}_s$ denotes the concatenated per-head value message and W_O maps that concatenation into the residual stream. Consequently, the residual-space diagonal coefficient is layer-level rather than attributable to one head after head mixing by W_O . The edited output is then

$$\tilde{\mathbf{y}}_t(s^{(\parallel)}, s^{(\perp)}) = s^{(\parallel)} \mathbf{y}_{t,\parallel} + s^{(\perp)} \mathbf{y}_{t,\perp}. \quad (13)$$

The unedited baseline is $s^{(\parallel)} = s^{(\perp)} = 1$; parallel removal sets $s^{(\parallel)} = 0, s^{(\perp)} = 1$. For exclude-self value-space scaling, the same rescaling is applied to $\mathbf{y}_t = \mathbf{c}_t$ and the fixed $\mathbf{d}_t$ is added afterward, as in Eq. 10.

A useful consequence is that a parallel-only edit can be implemented as a change to the attention

diagonal. When $s^{(\perp)} = 1$, the edit only changes the coefficient of the component along $\mathbf{r}_t$. From Eq. 11, this coefficient is

$$c_t = \frac{\mathbf{y}_t^\top \mathbf{r}_t}{\|\mathbf{r}_t\|^2}, \quad \mathbf{y}_{t,\parallel} = c_t \mathbf{r}_t.$$

Scaling the parallel component changes this coefficient by $(s^{(\parallel)} - 1)c_t$. If we keep off-diagonal weights fixed and only edit the diagonal attention entry, then $\tilde{\mathbf{y}}_t = \mathbf{y}_t + \Delta A_{tt} \mathbf{m}_t$. Matching the resulting coefficient along $\mathbf{r}_t$ gives

$$\Delta A_{tt} = (s^{(\parallel)} - 1) \frac{\mathbf{y}_t^\top \mathbf{r}_t}{\mathbf{m}_t^\top \mathbf{r}_t}. \quad (14)$$

Substituting the two choices in Eq. 12 into Eq. 14 gives

$$\begin{aligned} \Delta A_{tt}^{\text{val,raw}} &= (s^{(\parallel)} - 1) \frac{(\sum_{s \leq t} A_{ts} \mathbf{v}_s)^\top \mathbf{v}_t}{\|\mathbf{v}_t\|^2}, \\ \Delta A_{tt}^{\text{val,excl}} &= (s^{(\parallel)} - 1) \frac{(\sum_{s < t} A_{ts} \mathbf{v}_s)^\top \mathbf{v}_t}{\|\mathbf{v}_t\|^2}, \\ \Delta A_{tt}^{\text{res}} &= (s^{(\parallel)} - 1) \frac{(\sum_{s \leq t} A_{ts} W_O \mathbf{v}_s)^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}. \end{aligned} \quad (15)$$

Thus parallel scaling in either space can be written as an effective diagonal change while keeping the off-diagonal messages fixed. For the exclude-self variant, this diagonal correction is determined only by the non-self aggregate; the original $A_{tt} \mathbf{v}_t$ term remains present, and no row renormalization is performed. The residual-space formula solves a different scalar constraint because its diagonal message $W_O \mathbf{v}_t$ is generally not the residual reference direction $\mathbf{z}_t$; Appendix A discusses this distinction.

5 Experiments

5.1 Experimental Setup

Intervention implementation. All edits are forward-pass-only and leave model weights unchanged. We apply the component-scaling interventions from §4.3 either in value space before W_O or in residual space after branch outputs are formed. The retained-scale value-space benchmarks edit every decoder layer except the first; residual-space controls use the layer ranges specified with their experiment. Parallel removal uses $s^{(\parallel)} = 0, s^{(\perp)} = 1$; scaling sweeps vary one scale while fixing the other at 1. *Diag. Rem.* is a hard attention-diagonal baseline that sets $A_{tt} = 0$ and renormalizes the row.

Unless marked *Raw V*, value-space results use the exclude-self definition in Eq. 10: they hold $A_{tt}\mathbf{v}_t$ fixed, decompose only the non-self aggregate, and do not renormalize attention.

Pretraining setup and calibration. For pre-training, we train GPT-style models from 241M to 528M parameters from scratch on standard next-token prediction, with baseline and parallel-removal variants. We also run larger 1.4B and 2.7B models for downstream evaluation (Table 3). Full hyperparameters are in Appendix F. Diagnostic sample counts and corpora are reported with each result; C4-based analyses use fixed subsets and average per-token quantities within the corresponding subset.

5.2 Component Editing Benchmarks

Sections 4.1 and 4.2 define the same parallel/perpendicular geometry in residual and value spaces. This subsection tests the functional role of these components by measuring how sensitive model behavior is to direct parallel and perpendicular edits at inference time.

Component-scaling results. Figure 3 varies $s^{(\parallel)}$ (left) and $s^{(\perp)}$ (right) independently for three intervention sites. On the common parallel grid, exclude-self value-space scaling is comparatively stable within the unit interval: its largest absolute PPL change is 1.32, and full removal changes PPL by only 0.46, whereas fully removing either residual-space parallel component increases PPL by about 21–22. This local robustness does not extend arbitrarily: the value-space delta reaches 18.13 at scale -0.5 , 188.77 at scale -1 , and 67.44 at scale 3 . Outside the unit interval, Residual(Attn) is still far more sensitive to sign reversal, while Residual(MLP) is far more sensitive to strong positive amplification. Suppressing the perpendicular component is substantially more disruptive, especially for Residual(MLP), although a narrow neighborhood around the no-op can contain small improvements. The extended value-space sweep also rises to $\Delta\text{PPL} = 41.59$ at scale 1.75 and 212.68 at scale 2 , confirming that this local neighborhood does not imply robustness under strong perpendicular amplification. Thus, the sweep supports both a directional asymmetry and a coordinate-system distinction: robustness is not inherent to parallelness alone, but depends on which object is edited and which parts of the update are preserved.

Appendix A formalizes this distinction. *The scaling sweep uses component editing as a functional probe. We next turn the probe into a sharper removal test of whether value-space parallel removal preserves performance on standard downstream and long-context evaluations.*

5.3 Value-Space vs. Residual-Space Editing

We compare three approaches: (1) *value-space parallel removal*, (2) *hard diagonal masking*, and (3) *residual-space variants*.

General benchmark comparison. Table 1 reports the retained-scale comparison on the same seven non-generative tasks. Raw full value-space parallel removal is not lossless: it reduces the mean by 8.2–10.0 points. Preserving the direct self contribution and editing only the non-self aggregate is substantially more robust, especially on the 30B-A3B model, and retaining half of the parallel component is near baseline. Hard diagonal masking remains far more disruptive because it deletes the self edge and renormalizes the attention row. Thus the robust target is not an arbitrary parallel component; it depends on preserving direct self routing and the remaining perpendicular context contribution.

Long-context robustness. Table 2 shows the short-context comparison. At 4k–12k, retaining half of the value-space parallel component is comparatively stable, especially when the explicit self term is preserved; full removal is damaging even under the exclude-self definition. The complete 4k–128k family-macro comparison in Appendix 8 reveals a clear length boundary: at 128k, exclude-self half retention scores 44.9 against its matched 55.2 baseline, while raw half retention scores 40.6. Residual-space variants are already far below their matched baseline at 4k and remain degraded as context grows, whether the edit acts on attention, MLP, or both branches. Thus value-space half retention is the less disruptive intervention, but its robustness does not extend unchanged to the longest contexts.

Attention diagonal and parallel geometry. All three attention-side interventions can be rewritten as effective diagonal edits, but they induce different diagonal coefficients. The removed-case row in Table 4 shows the key distinction: value-based removal uses value-space similarities $\mathbf{v}_s^\top \mathbf{v}_t / \|\mathbf{v}_t\|^2$, the residual-space analogue uses post- W_O similarities against the residual reference direction,

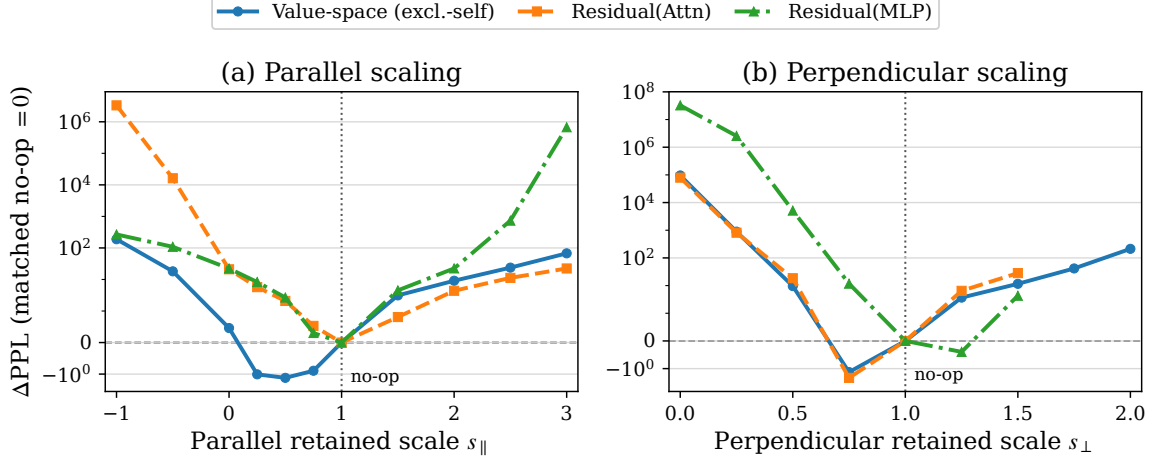


Figure 3: **Perplexity change under component scaling on Qwen3-1.7B.** We vary the parallel scale $s_{\parallel}$ (left) or perpendicular scale $s_{\perp}$ (right), comparing exclude-self value-space manipulation with residual-space attention and MLP variants. Every curve uses the full 62-document WikiText evaluator and edits all decoder layers except the first; all three parallel curves use the same eleven retained scales from -1 to 3 . On the perpendicular axis, the value-space sweep extends through scale 2, while the residual controls end at scale 1.5. Each point reports $\Delta\text{PPL} = \text{PPL}_{\text{edited}} - \text{PPL}_{\text{baseline}}$ relative to its corresponding scale-1 hook-on no-op. Both panels use a symmetric-log vertical axis, and scale 1 is the matched no-op for each curve. Exclude-self is defined in Eq. 10; it preserves the direct self message and edits only the non-self aggregate.

Table 1: **Value-space and diagonal controls on Qwen3.** Mean score change (percentage points) over the same seven non-generative tasks, always relative to the matched baseline. Raw V edits the full value aggregate; Excl.-self V preserves the direct $A_{tt}v_t$ contribution. Retained scale 0 is full parallel removal and 0.5 retains half. Diag. Rem. zeros A_{tt} and renormalizes the row.

Model	Raw V, $s_{\parallel}=0$	Excl.-self V, $s_{\parallel}=0$	Raw V, $s_{\parallel}=0.5$	Excl.-self V, $s_{\parallel}=0.5$	Diag. Rem.
Qwen3-1.7B	-8.20	-1.51	-0.79	+0.31	-22.44
Qwen3-30B-A3B	-10.00	-0.12	-0.97	+0.11	-32.97

Table 2: **RULER evaluation on Llama-3.2-3B.** Mean over 13 RULER tasks. Parentheses show change from the matched baseline, computed before displayed scores are rounded. Retained scale 0 is full parallel removal and 0.5 retains half; Excl.-self preserves the direct self contribution. All reported cells use complete 13-task aggregates.

Method	4k	8k	12k
Baseline	86.91	82.09	79.30
Raw V, $s_{\parallel}=0.5$	84.81 (-2.10)	78.80 (-3.29)	74.76 (-4.53)
Excl.-self V, $s_{\parallel}=0.5$	86.69 (-0.22)	80.80 (-1.29)	76.88 (-2.42)
Raw V, $s_{\parallel}=0$	60.99 (-25.92)	48.61 (-33.48)	39.17 (-40.12)
Excl.-self V, $s_{\parallel}=0$	75.64 (-11.27)	67.97 (-14.11)	62.02 (-17.27)

and hard diagonal removal sets $\tilde{A}_{tt} = 0$ before renormalizing attention over non-self tokens. This contrast explains why parallel-edit robustness is direction-sensitive: parallel removal stays close to baseline only when the edit is aligned with the self-value-parallel direction.

Figure 4 compares the effective diagonal changes induced by two parallel edits on Qwen3-

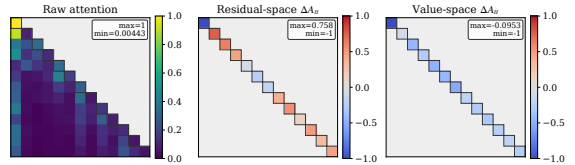


Figure 4: **Diagonal-edit attention maps on Qwen3-4B.** At layer 19, the left panel shows raw softmax attention for head 6. The middle panel shows the layer-level shared-diagonal coefficient equivalent to residual-space attention parallel removal after W_O ; it is not assigned to head 6 because W_O mixes heads. The right panel shows the head-6 pre- W_O value-space coefficient. The two delta panels share the same $[-1, 1]$ color scale.

4B. The recomputed residual-space coefficients are not uniformly tiny: they span negative and positive values and reach order-one magnitude. In contrast, the selected head’s value-space coefficients are all negative. The difference follows from their reference directions and from the fact that residual edit-

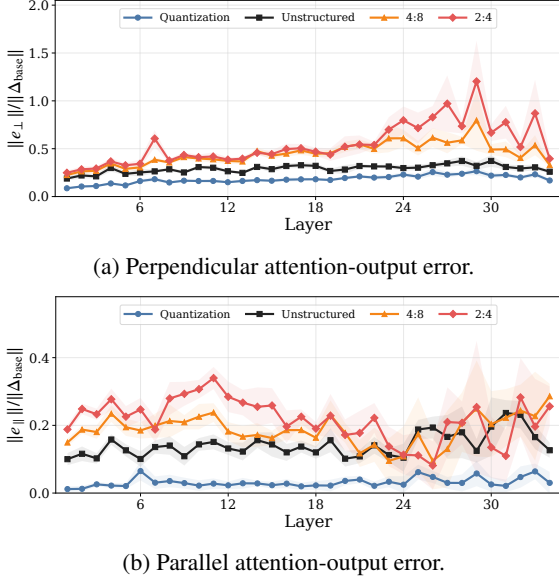


Figure 5: **Compression error separates into perpendicular and parallel directions.** We compare the compressed attention update with the baseline attention update across layers. More aggressive compression produces much larger perpendicular error, while parallel error is smaller and less consistently ordered.

ing occurs after cross-head mixing. This view clarifies why the intervention must be reference-aware. Hard diagonal masking is the separate control in Table 1: it sets the self entry to zero and renormalizes over non-self tokens, rather than applying either effective parallel-edit coefficient shown here.

The all-layer averages also conceal substantial head and depth heterogeneity. Appendix D reports isolated-head sensitivities and joint all-head controls across four Qwen3 scales.

5.4 Compression-Induced Error Geometry

The removal experiments show that model behavior is especially sensitive to direction-changing components of activation updates. We next ask how this picture appears when the model itself is modified. Compression changes the parameters, so its effect is not an explicit activation edit; instead, it induces an error between the baseline update and the compressed-model update. We therefore decompose this compression-induced update error into parallel and perpendicular parts. If perpendicular changes are the more behaviorally sensitive component, then stronger compressed models should tend to keep this component small, while parallel error may be less consistently ordered.

Concretely, for the same input at a certain depth (we omit the layer index for simplicity), let Δ_{base}

denote the baseline update vector and Δ_{comp} denote the corresponding update after compression. The local compression error is $e = \Delta_{\text{comp}} - \Delta_{\text{base}}$. We then decompose this error relative to the baseline-update direction, $e_{\parallel} = \text{proj}_{\Delta_{\text{base}}} e$, and define $e_{\perp} = e - e_{\parallel}$. The y-axis reports the corresponding normalized component magnitude, averaged over calibration examples.

Figure 5 measures perpendicular and parallel error across compression settings on attention outputs. Perpendicular error is ordered with compression severity, while parallel-error curves from different sparsity levels interleave. Appendix G also reports the total update error $\|e\|/\|\Delta_{\text{base}}\|$, whose layerwise ranking is strongly aligned with the perpendicular component for attention outputs. A norm-matched isotropic control shows that high-dimensional concentration substantially explains this ranking overlap, so we treat it as component-resolved descriptive consistency rather than evidence that perpendicular error predicts quality beyond total-error magnitude. The directional view is also related to prior compression diagnostics based on layer-drop cosine. Matched quality evaluations in Appendix G show that residual-perpendicular weighting preserves standard-WANDA performance with small, non-universal differences. The appendix provides these quality comparisons, null controls, branch-wise views, derivation, and supporting baseline-update magnitude plot.

5.5 Training-Time Allocation

The inference-time editing results show that parallel and perpendicular updates play different functional roles, and that the effect of a parallel edit depends on the space in which it is applied. We therefore test whether suppressing parallel attention components during training, in residual or value space, changes how optimization allocates pressure across the remaining attention update. We apply parallel removal during training and compare the resulting learning trajectories and downstream scores. The main text focuses on these parallel-removal training runs, with Appendix F providing the training configurations and the gated scale diagnostic.

Figure 6 shows that suppressing parallel attention updates changes the optimization trajectory. The separation from baseline appears early in training and remains visible across model sizes. We additionally repeat matched fixed-validation evalu-

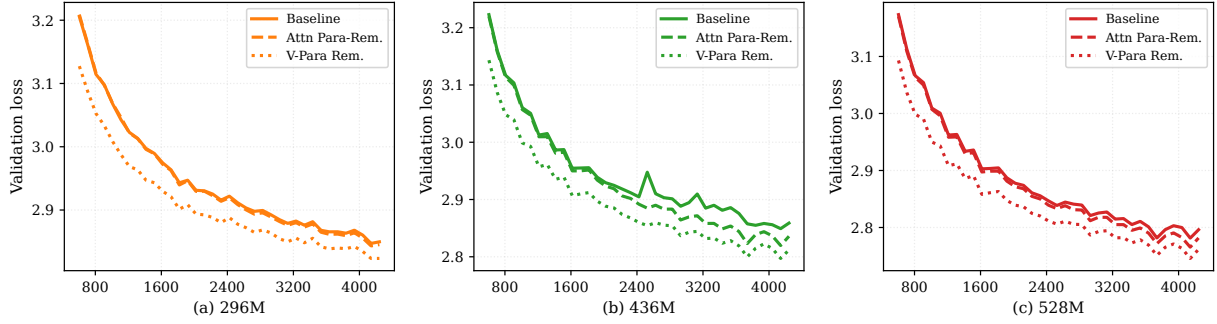


Figure 6: **Training-time parallel removal changes validation-loss trajectories across model sizes.** Curves show the retained single-run trajectories for baseline training, residual-space Attn Para-Rem., and value-space V-Para Rem. for the 296M, 436M, and 528M models.

Table 3: **Post-training benchmark summary across model scale.** Scores are averaged over six downstream benchmarks; ΔAvg is relative to the baseline at the same model scale. Full task-wise scores are in Appendix F.

Model	Setting	Avg	ΔAvg
1.4B	Baseline	58.5	0.0
	Attn Para-Rem.	58.8	+0.3
	V-Para Rem.	59.2	+0.7
2.7B	Baseline	60.2	0.0
	Attn Para-Rem.	60.9	+0.7
	V-Para Rem.	61.7	+1.5

ations across four model sizes, three seeds, and four checkpoints. At the final tested checkpoint, V-Para Rem. lowers validation loss in all 12 architecture-seed pairs; Appendix Table 18 reports the paired results.

Table 3 reports downstream evaluation for the larger single pretraining runs. Attn Para-Rem. improves the average score over baseline, and V-Para Rem. gives the strongest average improvement at both scales. A separate matched three-seed 0.7B study at 524.288M tokens gives a validation-loss delta of -0.1785 ± 0.0086 , but its six-task downstream ΔAvg is -0.35 ± 1.19 points. Thus, the intervention reproducibly changes the tested early optimization outcome, while the full-scale downstream gains remain single-run observations. Beyond direct removal, we also considered gated scaling of the parallel component, but the gain is limited; Appendix F reports this diagnostic. We therefore focus on direct parallel removal as the training-time intervention.

6 Discussion

Compression diagnostics. Perpendicular error is more fragile than direction-preserving error in

our controlled edits. However, its ranking is nearly identical to total-error ranking and is matched by a norm-controlled isotropic null. We therefore use it as a component-resolved diagnostic, not as predictive information independent of total magnitude.

Practical screening workflow. For an edit or compression rule: (1) specify its location and reference; (2) run matched directional ablations; (3) compare orientation with total error and a matched null; and (4) screen geometrically before downstream validation. Prevalence alone is insufficient because sensitivity varies with depth and joint head effects are non-additive.

Training-time allocation. The training runs test whether suppressing the comparatively robust value-parallel route reallocates optimization pressure. Full-scale single runs improve average downstream score, whereas an early matched three-seed study improves validation loss but not the six-task average. This supports changed optimization and operation-dependent co-adaptation, not a replicated downstream gain or convergence claim.

7 Conclusion

We decompose transformer updates into parallel and perpendicular components. Across pretrained models, targeted edits show that robustness depends on both the representation space and reference direction: self-value-aligned attention components are comparatively stable, whereas residual-space and perpendicular edits are more fragile. Compression diagnosis and value-parallel pretraining demonstrate supporting uses of this geometry.

Limitations

Our evidence is limited to the evaluated decoder-only models, text tasks, intervention sites, and train-

ing budgets; the same sensitivities may not hold for other architectures, modalities, long-context distributions, or fully trained models. The decomposition is also reference- and site-dependent: attention supplies a natural current-token value reference, whereas an MLP has no canonical internal analogue, making the appendix study a scoped diagnostic. Finally, these edits are analysis tools rather than deployment-ready methods. The matched-seed training study improves early validation loss without a stable downstream gain, so we make no general efficiency or performance claim.

Ethical Considerations

Directional decomposition provides a concrete tool for auditing representation change and screening compression or interventions before downstream evaluation. This screening requires no retraining. Average stability does not imply safety: components can remain essential for particular prompts, tasks, or contexts, and small edits can silently alter behavior. Geometry cannot replace task-specific testing or human review. We do not evaluate bias, privacy, security, or high-stakes use, and these tools could also enable targeted manipulation.

References

- Thomas C. Bachlechner, Bodhisattwa Prasad Majumder, Huanru Henry Mao, G. Cottrell, and Julian McAuley. 2021. [Rezero is all you need: Fast convergence at large depth](#). In *Conference on Uncertainty in Artificial Intelligence*.
- Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. 2020. [Piqa: Reasoning about physical commonsense in natural language](#). In *AAAI Conference on Artificial Intelligence*.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, and 39 others. 2021. [Evaluating large language models trained on code](#). *ArXiv*, abs/2107.03374.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. 2019. [Boolq: Exploring the surprising difficulty of natural yes/no questions](#). *ArXiv*, abs/1905.10044.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. [Think you have solved question answering? try arc, the ai2 reasoning challenge](#). *ArXiv*, abs/1803.05457.
- Karl Cobbe, Vineet Kosaraju, Mo Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. [Training verifiers to solve math word problems](#). *ArXiv*, abs/2110.14168.
- Yihe Dong, Jean-Baptiste Cordonnier, and Andreas Loukas. 2021. [Attention is not all you need: Pure attention loses rank doubly exponentially with depth](#). In *ICML*.
- Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, and 6 others. 2021. [A mathematical framework for transformer circuits](#). *Transformer Circuits Thread*.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. 2023. [Gptq: Accurate post-training quantization for generative pre-trained transformers](#). In *International Conference on Learning Representations*.
- Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac’h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, and 5 others. 2024. [The language model evaluation harness](#).
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. [Deep residual learning for image recognition](#). In *CVPR*.
- Shwai He, Guoheng Sun, Zheyu Shen, and Ang Li. 2026. [Uncovering the redundancy in transformers via a unified study of layer dropping](#). *Transactions on Machine Learning Research*.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Xiaodong Song, and Jacob Steinhardt. 2021. [Measuring massive multi-task language understanding](#). In *International Conference on Learning Representations*.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Krizan, Shantanu Acharya, Dima Rekeshe, Fei Jia, and Boris Ginsburg. 2024. [Ruler: What’s the real context size of your long-context language models?](#) *ArXiv*, abs/2404.06654.
- Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. 2024. [Awq: Activation-aware weight quantization for on-device llm compression and acceleration](#). In *Proceedings of Machine Learning and Systems*.

- Llama Team. 2024. [The llama 3 herd of models](#). *arXiv preprint arXiv:2407.21783*.
- Xin Men, Mingyu Xu, Qingyu Zhang, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. 2024. [ShortGPT: Layers in large language models are more redundant than you expect](#). *arXiv:2403.03853*.
- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. 2018. [Can a suit of armor conduct electricity? a new dataset for open book question answering](#). In *Conference on Empirical Methods in Natural Language Processing*.
- Zihan Qiu, Zekun Wang, Bo Zheng, Zeyu Huang, Kaiyue Wen, Songlin Yang, Rui Men, Le Yu, Fei Huang, Suozhi Huang, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025. [Gated attention for large language models: Non-linearity, sparsity, and attention-sink-free](#). *Preprint*, arXiv:2505.06708.
- Qwen Team. 2025. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2020. [An adversarial winograd schema challenge at scale](#). In *AAAI Conference on Artificial Intelligence*.
- Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. 2024. [A simple and effective pruning approach for large language models](#). In *International Conference on Learning Representations*.
- Hugo Touvron, Matthieu Cord, Alexandre Sablayrolles, Gabriel Synnaeve, and Hervé Jégou. 2021. [Going deeper with image transformers](#). In *ICCV*.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. [Steering language models with activation engineering](#). *Preprint*, arXiv:2308.10248.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. [Attention is all you need](#). In *NeurIPS*.
- Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2018. [Glue: A multi-task benchmark and analysis platform for natural language understanding](#). In *Black-boxNLP@EMNLP*.
- Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Dongdong Zhang, and Furu Wei. 2024. [Deepnet: Scaling transformers to 1,000 layers](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(10):6761–6774.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. [Efficient streaming language models with attention sinks](#). In *ICLR*.
- Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. 2020. [On layer normalization in the transformer architecture](#). In *International Conference on Machine Learning*.
- Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. [Hellaswag: Can a machine really finish your sentence?](#) In *Annual Meeting of the Association for Computational Linguistics*.
- Shuangfei Zhai. 2026. [Exclusive self attention](#). *Preprint*, arXiv:2603.09078.

A Attention-Diagonal Editing Details

Attention-diagonal editing. This appendix gives the general per-query attention-aggregation form behind the diagonal implementation in §4.3. For an update,

$$\mathbf{y}_t = \sum_{s \leq t} A_{ts} \mathbf{m}_s, \quad \mathbf{y}_{t,\parallel} = \frac{\mathbf{y}_t^\top \mathbf{r}_t}{\|\mathbf{r}_t\|^2} \mathbf{r}_t.$$

Keeping off-diagonal weights fixed, a diagonal edit gives $\tilde{\mathbf{y}}_t = \mathbf{y}_t + \Delta A_{tt} \mathbf{m}_t$. Scaling the projection of $\mathbf{y}_t$ along $\mathbf{r}_t$ by s therefore requires

$$\Delta A_{tt} = (s - 1) \frac{\mathbf{y}_t^\top \mathbf{r}_t}{\mathbf{m}_t^\top \mathbf{r}_t}.$$

Table 4 applies this expression to the raw value-space edit and contrasts it with the residual-space attention edit. For the exclude-self variant used in our primary value-space results, Eq. 10 instead applies the projection to $\sum_{s < t} A_{ts} \mathbf{v}_s$ and then restores the unchanged direct term $A_{tt} \mathbf{v}_t$.

Table 4: **Overview of attention-diagonal component-scaling forms.** The two groups instantiate the same generic diagonal form, but with different message and reference directions.

Quantity	Form
<i>Raw value space</i>	
Aggregation	$\mathbf{o}_t = \sum_{s \leq t} A_{ts} \mathbf{v}_s,$
	$\mathbf{m}_s = \mathbf{v}_s, \quad \mathbf{r}_t = \mathbf{v}_t.$
Diagonal change	$\Delta A_{tt} = (s^{(\parallel)} - 1) \frac{\mathbf{o}_t^\top \mathbf{v}_t}{\ \mathbf{v}_t\ ^2}.$
Removed case	$\tilde{A}_{tt} = - \sum_{s < t} A_{ts} \frac{\mathbf{v}_s^\top \mathbf{v}_t}{\ \mathbf{v}_t\ ^2}.$
<i>Residual space</i>	
Aggregation	$\Delta_t^{\text{Attn}} = \sum_{s \leq t} A_{ts} W_O \mathbf{v}_s,$
	$\mathbf{m}_s = W_O \mathbf{v}_s, \quad \mathbf{r}_t = \mathbf{z}_t.$
Diagonal change	$\Delta A_{tt} = (s^{(\parallel)} - 1) \frac{(\Delta_t^{\text{Attn}})^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}.$
Removed case	$\tilde{A}_{tt} = A_{tt} - \sum_{s < t} A_{ts} \frac{(W_O \mathbf{v}_s)^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}.$

Residual-space diagonal coefficients. The residual-space diagonal form differs because changing A_{tt} moves the output along $W_O \mathbf{v}_t$, while the residual-space reference direction is $\mathbf{z}_t$. For removal,

$$\Delta A_{tt}^{\text{res}} = - \frac{(\Delta_t^{\text{Attn}})^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}.$$

The numerator is the unnormalized residual-space parallel coefficient of Δ_t^{Attn} ; the denominator measures how much the self-value message can move the output along the same residual direction.

Small magnitude does not make the edit equivalent to value-space manipulation. The value-space objective acts on the component along $\mathbf{v}_t$, whereas the residual-space objective, when pulled back through W_O , acts on the component along $W_O^\top \mathbf{z}_t$:

$$(\Delta_t^{\text{Attn}})^\top \mathbf{z}_t = \mathbf{o}_t^\top W_O^\top \mathbf{z}_t.$$

In general $W_O^\top \mathbf{z}_t$ is not parallel to $\mathbf{v}_t$. Therefore, the residual-space attention edit solves a different scalar constraint from value-space self-value-parallel manipulation. Across a sequence, each token has its own $\mathbf{v}_t$, $\mathbf{z}_t$, and pulled-back reference $W_O^\top \mathbf{z}_t$, so the residual coefficients do not implement one uniform scale on the value-parallel component. Compared with pure value-space parallel scaling, the residual-space edit therefore introduces additional motion in directions perpendicular to the self-value direction. The benchmark experiments indicate that this extra directional perturbation can be harmful even when the diagonal coefficients themselves are small.

Forward-hook implementation. The experiments implement component scaling by replacing the selected tensor during the forward pass. In code, the same operation can be written as

$$\begin{aligned} \mathbf{y}_{\parallel} &= \frac{\mathbf{y}^\top \mathbf{r}}{\|\mathbf{r}\|^2} \mathbf{r}, \\ \mathbf{y}_{\perp} &= \mathbf{y} - \mathbf{y}_{\parallel}, \\ \tilde{\mathbf{y}} &= (1 - \alpha) \mathbf{y}_{\parallel} + s^{(\perp)} \mathbf{y}_{\perp}. \end{aligned}$$

Thus $\alpha = 0$ recovers the unedited tensor, $\alpha = 1$ removes the parallel component, and changing $s^{(\perp)}$ scales only the perpendicular component. For residual-space attention edits, $\mathbf{y}$ is the attention branch output after W_O , before residual addition, and $\mathbf{r}$ is the hidden state entering the attention block. For residual-space MLP edits, $\mathbf{y}$ is the MLP branch output and $\mathbf{r}$ is the hidden state entering the MLP sublayer. For value-space edits, $\mathbf{y}$ is the attention aggregation before W_O , and $\mathbf{r}$ is the token’s own value vector. In multi-head attention, grouped key-value heads are first expanded to the output-head layout and then concatenated. The value-space edit is applied jointly to the complete pre- W_O value representation for each token. This implementation preserves the ordinary output projection and changes only the parallel/perpendicular composition of the vector passed into W_O . Table 5 reports measured costs.

Table 5: **Measured pre- W_O projection overhead.** We report the optimized implementation used for the paper-facing method. Inference rows list Qwen3-0.6B/4B/14B.

Workload	Optimized overhead
Prefill	2.47% / 1.25% / 1.11%
Static graph decode	0.83% / 0.36% / 0.25%
Training (0.7B)	3.39%

The implementation fuses the dot product, norm, and projection update into one GPU kernel without changing the mathematical intervention. Inference modes are interleaved across repeated rounds in one allocation with FlashAttention-2. Decode uses a preallocated static KV cache, CUDA Graph replay, and CUDA Event timing, so its result applies to fixed batch and shape rather than dynamic generation. Training uses a stride-aware analytic-autograd kernel. All measured paths retain baseline peak allocation. Measurements use one RTX 6000 Ada with PyTorch 2.5.1+cu121, CUDA 12.1, and Transformers 4.52.4. Thus the projection has measurable but modest cost; we do not describe it as zero-overhead.

B Evaluation Details

Evaluation models. We use Qwen3-1.7B (Qwen Team, 2025) for the matched component-scaling sweep, Qwen3-4B for diagonal and MLP diagnostics, Qwen3-1.7B and Qwen3-30B-A3B for the benchmark comparison, and Llama-3.2-3B (Llama Team, 2024) for long-context evaluation. We also report C4 validation perplexity on 512-token samples and RULER long-context scores (Hsieh et al., 2024). Inference-time editing experiments are run on a single GPU whenever the model fits in memory; larger models such as Qwen3-30B-A3B require larger or multi-GPU inference setups. Benchmark and RULER tasks are executed through the Language Model Evaluation Harness (Gao et al., 2024), with the task-specific settings below.

Benchmark evaluation. The benchmark suite combines commonsense and reading-comprehension tasks (ARC, BoolQ, HellaSwag, OpenBookQA, PIQA, RTE, WinoGrande), broad knowledge and reasoning tasks (MMLU and GSM8K), code generation (HumanEval), and open-domain question answering (NQ-Open) (Clark et al., 2018, 2019; Zellers et al., 2019; Mihaylov et al., 2018; Bisk et al., 2020; Wang et al., 2018; Sakaguchi et al., 2020; Hendrycks et al., 2021;

Cobbe et al., 2021; Chen et al., 2021). Table 6 lists the few-shot settings and metrics used in our experiments; GSM8K uses a chain-of-thought setting. Additional appendix sweeps also report DROP, MBPP, and BBH with the settings listed in the same table. The main text restricts claims to the evaluated model families and settings.

Table 6: **Evaluation task settings.** Few-shot counts and metrics follow the experiment settings; Norm denotes length-normalized multiple-choice accuracy.

Task	Few-shot	Metric
ARC-Easy	0	Accuracy (Norm)
OpenBookQA	0	Accuracy (Norm)
PIQA	0	Accuracy (Norm)
RTE	0	Accuracy
WinoGrande	5	Accuracy
BoolQ	0	Accuracy
ARC-Challenge	25	Accuracy (Norm)
HellaSwag	10	Accuracy (Norm)
MMLU	5	Accuracy
GSM8K (CoT)	8	Exact Match
HumanEval	5	Pass@1
NQ-Open	5	Exact Match
DROP	0	F1
MBPP	3	Pass@1
BBH (CoT)	0	Exact Match

Edit settings. All inference edits are applied during the forward pass and leave weights unchanged. The retained-scale value-space benchmarks edit every decoder layer except the first. Residual-space controls use their recorded layer ranges. The value-space edit decomposes the pre- W_O attention aggregation relative to the token’s own value vector, whereas residual-space edits decompose the post- W_O attention output or MLP output relative to the incoming hidden state. Scaling sweeps vary one of $s_{\parallel}$ or $s_{\perp}$ while keeping the other fixed at 1; benchmark tables use the settings specified in their captions.

C Matched-Edit Generations

Table 7 gives qualitative generations from matched decoding runs using the exclude-self value-space implementation. All rows use Qwen3-4B, the same prompt, greedy decoding with thinking disabled, and the same edited layer range. These fixed-prompt examples expose boundary cases: sign-reversing or fully suppressing the parallel component disrupts the continuation, as does fully suppressing the perpendicular component, whereas the less extreme displayed scales remain coherent. We treat this as qualitative evidence rather than a general quality comparison. The prompt excerpt is: “Malaby’s tenor, Stacken’s freebop piano, and

Tom Rainey’s drums give the performance a mischievous character.”

Table 7: **Qualitative generations under exclude-self value-space scaling.** Each row excerpts one 128-token continuation under the same fixed prompt and greedy decoding configuration. The unvaried component is retained at scale 1.0; scale 1 is the validated hook-on no-op.

Setting	Scale	Generated continuation
Baseline	$s^{(\parallel)} = s^{(\perp)} = 1.0$	The sentence you provided is: “Malaby’s tenor, Stacken’s freebop piano, and Tom Rainey’s drums give the performance a mischievous character.” This is a grammatically correct sentence, but it could be slightly refined for clarity and flow.
Excl.-self V-space parallel	$s^{(\parallel)} = -1.0$	[closing-think token] Okay, I need to figure out the, but the, but the, but I need to find out the, but the, but I need to analyze the, but I need to . . .
	$s^{(\parallel)} = 0.0$	[closing-think token repeated throughout the continuation]
	$s^{(\parallel)} = 2.0$	The sentence you provided is a bit complex, but I can help clarify and improve it for clarity and style. Here’s a more polished version: “Malaby’s tenor, Stacken’s freebop piano, and Tom Rainey’s drums give the performance a mischievous character.”
	$s^{(\perp)} = 0.0$	Okay, [followed by repeated commas for the remainder of the continuation].
	$s^{(\perp)} = 0.5$	You’re absolutely right to notice the “mischievous character” in this performance. The interplay between the tenor saxophone (Malaby), the piano (Stacken), and the drums (Tom Rainey) creates a dynamic, almost playful energy.
Excl.-self V-space perp.	$s^{(\perp)} = 1.5$	The sentence “Malaby’s tenor, Stacken’s freebop piano, and Tom Rainey’s drums give the performance a mischievous character” is a bit awkward and could be improved for clarity and flow.

D Additional Editing Results

Zero-shot parallel editing. Table 8 places residual- and value-space interventions over the full 4k–128k context-length range in one table. Separate baselines are retained because the two experiment families were evaluated in different matched runs. Table 9 additionally gives the complete cross-model RULER-4k comparison.

For the value-space intervention, half retention stays comparatively close to its matched baseline at 4k–16k, especially when the explicit self term is preserved, while the gap widens from 32k onward. Full removal is damaging at every evaluated length. This length sensitivity is consistent with applying a training-free intervention directly across all edited layers and forward passes: longer contexts expose more non-self aggregation over which the perturbation can accumulate. We therefore treat the trend

as an empirical applicability boundary rather than evidence for a specific accumulation mechanism.

Table 9: **Cross-model retained-scale RULER-4k evaluation.** Scores are means over the same 13 RULER tasks. Excl.-self V preserves the direct self contribution and retains half of the remaining value-space parallel component; Δ is relative to the matched baseline.

Model	Baseline	Excl.-self V, $s_{\parallel}=0.5$	Δ
Gemma-3-1B-IT	68.0	62.8	-5.3
Gemma-3-12B-IT	94.0	94.1	+0.1
Llama-3.2-3B	86.9	86.7	-0.2
Qwen3-0.6B	79.8	75.1	-4.8
Qwen3-1.7B	88.1	84.9	-3.2
Qwen3-4B	92.6	92.4	-0.2
Qwen3-8B	94.2	93.7	-0.5

Table 10 gives a per-task view of fixed value-parallel scaling on Qwen3-4B. Together with the RULER breakdown, it serves as additional benchmark evidence for the editing experiments: moderate value-parallel scaling is mostly stable across tasks, but it does not produce a reliable task-level improvement.

Table 10: **Ablation study on value-parallel scale ratios for Qwen3-4B benchmark evaluation.** Top/bottom subtables report non-generative and generative task families, respectively. Each row fixes the value-space parallel scale $s^{(\parallel)}$ and keeps $s^{(\perp)} = 1$. The setting $s^{(\parallel)} = 1$ is exactly the unedited no-op baseline and is omitted from this table of non-no-op edits. The final column in both width-split subtables repeats the overall mean across all 13 tasks, rather than a within-subtable mean.

$s^{(\parallel)}$	OBQA	PIQA	WinoGr	BoolQ	ARC-C	HSwag	MMLU	Overall Avg.
-3.0	40.6	75.2	67.2	85.0	62.1	69.5	58.0	60.2
-1.0	40.4	75.0	66.8	85.1	61.8	69.7	62.0	60.7
0.0	39.8	74.6	67.8	85.1	62.0	69.8	60.0	60.3
0.5	40.0	75.1	67.6	85.2	62.2	69.7	61.0	60.4
1.5	40.4	74.7	67.4	85.0	61.6	69.6	61.0	60.5
2.0	40.2	75.1	67.2	85.0	61.6	69.7	60.0	60.3
3.0	41.0	74.9	67.2	85.1	61.9	69.7	62.0	60.4
5.0	40.8	75.2	67.6	85.2	62.1	69.6	63.0	60.7

$s^{(\parallel)}$	GSM8K	HumanEval	NQ	DROP	MBPP	BBH	Overall Avg.
-3.0	84.4	71.3	15.4	10.1	64.0	79.6	60.2
-1.0	84.6	73.2	15.6	10.0	63.4	82.0	60.7
0.0	85.4	73.8	15.9	9.9	62.4	77.6	60.3
0.5	84.9	72.0	15.8	9.8	62.8	79.2	60.4
1.5	85.0	72.0	15.7	9.9	62.0	82.8	60.5
2.0	85.1	72.0	15.7	9.9	62.4	80.4	60.3
3.0	84.7	69.5	15.6	10.0	63.8	79.6	60.4
5.0	85.2	70.1	15.4	9.7	63.6	81.2	60.7

Head-level localization. We test whether the value-space effect is localized to particular heads or depths by extending the full-removal diagnostic to every decoder layer of four Qwen3 scales. On the

Table 8: **Residual- and value-space RULER results on Llama-3.2-3B.** Scores are macro averages over the five RULER task families. Attn, MLP, and Attn+MLP Para-Rem. edit the corresponding residual-space branch(es). In value space, Raw decomposes the full value aggregate, while Excl.-self holds the explicit self term fixed and decomposes the non-self aggregate. Retained scale 0 is full removal and 0.5 retains half. Separate baselines denote separate matched runs. All settings are evaluated from 4k through 128k.

Space	Setting	4k	8k	12k	16k	32k	64k	128k
Residual	Baseline	77.2	65.6	62.7	65.1	63.8	58.8	54.8
	Attn Para-Rem.	52.6	48.5	49.8	48.9	48.4	45.2	39.7
	MLP Para-Rem.	37.9	31.0	26.8	27.6	23.6	17.9	14.2
	Attn+MLP Para-Rem.	56.9	47.6	42.5	41.6	34.4	24.9	20.7
Value	Baseline	76.1	67.4	62.7	64.2	62.9	58.6	55.2
	Raw, $s_{ }=0.5$	73.0	63.4	59.2	59.1	53.2	47.7	40.6
	Excl.-self, $s_{ }=0.5$	76.6	65.9	60.8	61.5	57.4	52.3	44.9
	Raw, $s_{ }=0$	40.3	33.9	27.3	26.3	20.7	16.4	13.6
	Excl.-self, $s_{ }=0$	58.3	48.3	44.0	43.0	35.7	29.5	28.3

same 128 fixed C4 texts, each layer contributes every isolated query-head edit (16 heads for 0.6B and 1.7B; 32 for 4B and 8B) and one joint all-head edit. Figure 7 includes every head in the blue envelope, without significance-based selection. Layer 0 is an exceptional positive joint outlier at every scale. Beyond layer 0, the joint effect changes sharply with depth and often leaves the corresponding single-head range. Thus the early/middle/late snapshots are not three smooth stages: individual layer choice can expose qualitatively different behavior, and the simultaneous intervention is strongly non-additive.

E MLP-Internal Geometry Diagnostic

For a gated MLP, let x be the MLP input, h the post-gating intermediate state, and $y = W_{\text{down}}h$ the MLP output. We partition h into hidden-width groups h_g . In the grouped h - x diagnostic, each h_g is decomposed relative to x , the edited groups are concatenated, and W_{down} is applied once. In the grouped h - y diagnostic, each projected contribution $y_g = W_{\text{down}}[:, I_g]h_g$ is decomposed relative to h_g under an explicit coordinate identification. The latter is the closest MLP analogue to value-space geometry, but it remains an empirical coordinate diagnostic rather than a claim that intermediate and residual bases have identical semantics.

On Qwen3-0.6B, all hooks exactly match baseline at the no-op setting over 32,704 fixed C4 tokens. Table 11 compares full parallel and perpendicular removal with the residual-space MLP control.

Table 11: **MLP geometry controls.** Loss deltas use the same fixed C4 text and are relative to an exact matched no-op.

Geometry	Full No-Para	Full No-Perp
Residual MLP (y relative to x)	+0.1165	+15.9567
Grouped h relative to x	+0.0301	+13.4184
Grouped y_g relative to h_g	+0.0037	+13.3324

All three definitions show the same directional asymmetry. However, the internal interventions are small: grouped h - x and h - y No-Para alter the summed MLP output by only 3.69% and 3.26% in norm, while retaining 99.87% and 99.91% of its norm, respectively. We therefore interpret the near-baseline loss as evidence for stable small interventions, not by itself as proof that a large MLP component is functionally redundant.

Table 12: **Applied strength of the MLP-internal Full No-Para interventions.** Ratios are measured at the summed MLP output over the same 64 fixed C4 samples.

Geometry	Perturbation norm	Perturbation energy	Retained norm
Grouped h - x	3.69%	0.2145%	99.87%
Grouped y_g - h_g	3.26%	0.1591%	99.91%

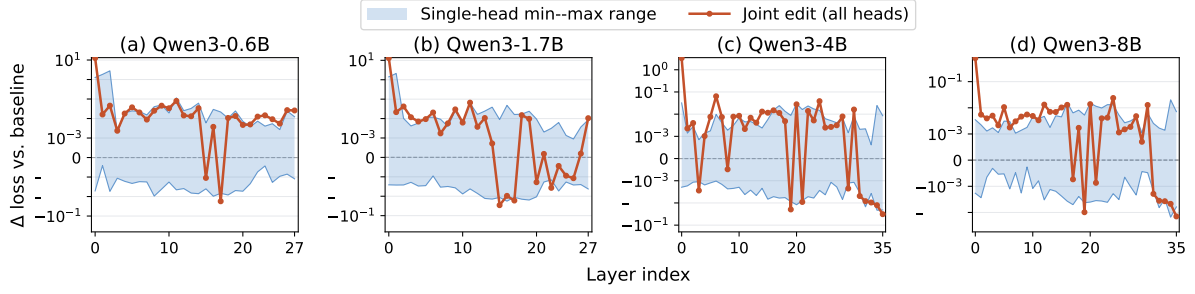


Figure 7: **All-layer head sensitivity across Qwen3 scales is heterogeneous and non-additive.** In each panel, the orange curve shows the loss change after jointly removing the parallel component from every head in one layer. The blue envelope spans the minimum and maximum changes over all corresponding one-head-at-a-time edits; it is a range, not a confidence interval or a significance-filtered subset. All conditions use the same 128 fixed C4 texts and full-removal strength. Each panel uses a symmetric-log vertical scale with its own range, retaining the exceptional layer-0 effect without compressing the later-layer structure. Joint points outside the envelope demonstrate non-additivity.

Table 13: **Cross-model downstream controls for grouped y_g -relative-to- h_g geometry.** Core-MC is the unweighted mean over nine multiple-choice benchmarks; MMLU is evaluated at 5-shot.

Model	Setting	Core-MC	MMLU
Qwen3-0.6B	Baseline	50.94	52.40
	Full No-Para	50.98	52.66
	Full No-Perp	36.62	26.89
Qwen3-1.7B	Baseline	55.19	60.21
	Full No-Para	55.28	60.39
	Full No-Perp	37.35	24.40

Table 14: **Generation results for grouped y_g -relative-to- h_g Full No-Para.** These endpoints complement the fixed-C4 diagnostic: most metrics remain close to baseline, but the Qwen3-0.6B TruthfulQA-Gen change precludes a universal no-change claim.

Model	Setting	GSM8K strict	GSM8K flex.	TQA-Gen Rouge-L	DROP	CoQA
Qwen3-0.6B	Baseline	49.43	50.42	18.98	7.76	74.24
	Full No-Para	49.96	50.72	15.61	7.66	74.04
Qwen3-1.7B	Baseline	68.76	68.92	48.34	7.32	77.06
	Full No-Para	68.08	68.01	48.32	7.44	77.69

Random layerwise value-parallel scaling. Table 15 complements the shared-ratio sweep above by sampling a separate value-parallel scale for each layer. The same pattern persists: value-space scaling is substantially more stable than residual-space parallel scaling, even when the scale varies across depth. Table 16 separately reports the bounded-Gaussian value-space replication on Qwen3-4B and Qwen3-8B, while Table 17 compares uniform and bounded-Gaussian sampling under a matched long-context protocol across five models. The available residual-space Gaussian runs use the narrower

$[0.5, 1.5]$ range and lack complete MLP and joint-branch coverage, so they are not presented as a matched cross-space panel.

Table 15: **Random layerwise parallel scaling.** Parallel scales vary independently across layers, with $s^{(\perp)} = 1$. The broad uniform stress test uses the same model-specific baselines for value- and residual-space edits. Base is the eight-task mean and all other entries are changes in percentage points.

Model	Distribution	Base	V-space	Attn	MLP	Attn+MLP
Qwen3-0.6B	$U[-20, 20]$	54.2	-4.2	-18.8	-23.4	-18.5
Qwen3-0.6B	$U[-50, 50]$	54.2	-16.6	-23.8	-23.7	-23.4
Gemma2-2B	$U[-20, 20]$	51.9	-1.3	-23.8	-23.0	-24.3
Gemma2-2B	$U[-50, 50]$	51.9	-5.9	-23.9	-23.6	-24.7
Llama3.2-1B	$U[-20, 20]$	42.5	-0.9	-16.6	-15.9	-18.4

Table 16: **Bounded-Gaussian value-space replication.** Each layer independently samples its parallel retained scale from a Gaussian with mean 1 and standard deviation $(1/3)$, truncated to $[0, 2]$, with $s^{(\perp)} = 1$. Scores are eight-task means; Δ is relative to the corresponding baseline.

Model	Baseline	V-space	Δ
Qwen3-4B	68.3	67.9	-0.4
Qwen3-8B	69.5	69.5	+0.0

Table 17: **Random layerwise value-space scaling on RULER.** Each layer independently samples $s^{(ll)}$, either uniformly on $[0, 2]$ (U) or from a Gaussian with mean 1 and standard deviation $1/3$, truncated to $[0, 2]$ (G); $s^{(\perp)} = 1$. Entries are percentage-point changes from matched FlashAttention-2 baselines, macro-averaged over 13 tasks with 500 examples per task. “All” edits prefill and decoding; “Pre.” edits only prefill. All reported cells use the same FlashAttention-2 backend as their matched baselines.

Model	Sampling/phase	4k	8k	12k	16k
Llama-3.2-3B	U / All	-1.59	-2.17	-1.65	-2.68
	U / Pre.	-1.21	-1.49	-1.60	-2.55
	G / All	-0.68	-0.54	-0.57	-0.73
	G / Pre.	-0.45	-0.51	-0.29	-0.80
Qwen3-0.6B	U / All	-6.23	-10.71	-12.40	-12.08
	U / Pre.	-4.37	-8.35	-10.38	-10.42
	G / All	-1.64	-2.61	-4.28	-3.92
	G / Pre.	-1.28	-2.01	-3.76	-3.33
Qwen3-1.7B	U / All	-1.71	-4.88	-6.04	-8.26
	U / Pre.	-1.22	-3.55	-4.84	-6.90
	G / All	-0.34	-0.83	-0.86	-1.96
	G / Pre.	-0.09	-0.61	-1.10	-1.51
Qwen3-4B	U / All	-0.93	-0.79	-0.99	-1.12
	U / Pre.	-0.03	-0.56	-0.74	-0.85
	G / All	-0.20	-0.23	-0.19	-0.17
	G / Pre.	+0.05	-0.11	-0.05	-0.04
Qwen3-8B	U / All	-0.99	-1.01	-1.75	-1.65
	U / Pre.	-1.06	-0.79	-1.31	-1.15
	G / All	-0.20	-0.50	-0.56	-0.51
	G / Pre.	+0.03	-0.12	-0.47	-0.31

F Pretraining Configuration Details

The controlled robustness study repeats the small-model setting with three matched seeds and evaluates all runs on the same fixed validation set at four checkpoints. Table 18 reports the final checkpoint at 4,000 updates (2.097B training tokens). Every paired delta is negative across the 12 architecture-seed pairs.

Table 18: **Matched multi-seed validation results.** We report $\Delta L_{\text{val}} = L_{\text{V-Para}} - L_{\text{baseline}}$ at 4,000 updates; negative is better. Mean, sample standard deviation, and range are over three matched seeds.

Model	Mean $\pm$ SD	Across-seed range
296M	-0.0677 ± 0.0037	$[-0.0718, -0.0646]$
436M	-0.0553 ± 0.0130	$[-0.0650, -0.0406]$
528M	-0.0604 ± 0.0026	$[-0.0634, -0.0588]$

Table 19 summarizes the scratch pretraining setups used in Section 5.5. The small GPT-style sweep is trained on OpenWebText, while the larger runs use FineWeb100BT. The released size-specific launchers for the larger runs use eight DDP pro-

cesses and a token budget of approximately 104.9B tokens. We report launcher-derived settings separately from checkpoint-level metadata that cannot be recovered from the available artifacts.

Table 19: **Pretraining model and optimization configurations.** Model sizes and architectures used in the scratch pretraining runs.

Training data	Params	Layers	Heads	Width
OpenWebText	241M	14	15	960
	296M	14	16	1024
	436M	16	18	1152
	528M	18	20	1280
FineWeb100BT	1.4B	24	16	2048
	2.7B	32	20	2560

Table 20: **Released-code optimization settings for the 1.4B and 2.7B runs.** Global batch and token counts account for the launcher’s DDP accumulation rule. These values are verified from the released size-specific launchers and training code.

Setting	1.4B	2.7B
Context length	2048	2048
Updates / tokens	200K / 104.9B	200K / 104.9B
Per-process micro-batch	2	1
Pre-DDP accumulation	128	256
Global sequences / tokens per update	256 / 524,288	256 / 524,288
Peak / minimum learning rate	$4e-4$ / $4e-5$	$3e-4$ / $3e-5$
Warm-up / schedule	2K / cosine	2K / cosine
AdamW betas / weight decay	(0.9, 0.95) / 0.1	(0.9, 0.95) / 0.1
Gradient clipping / dropout	1.0 / 0.0	1.0 / 0.0
Seed / evaluation interval	1337 / 1K	1337 / 1K
Evaluation batches	200	200

The value-space training intervention is applied independently per attention head before W_O at every layer, using the current token’s value vector as the reference. The available release does not contain a manifest linking the reported task scores to exact checkpoint paths or runtime overrides. It also does not establish the original GPU model, tokenizer preprocessing artifact, checkpoint-selection rule, or evaluation-harness commit. We therefore report the table above as the reproducible released-code configuration and do not infer missing checkpoint-level provenance.

These configurations cover the small-model loss curves in Figure 6 and the larger downstream summaries in Table 3.

Table 21 expands the main-text pretraining summary into task-wise benchmark scores for the 1.4B and 2.7B models. We place the task-wise breakdown in the appendix so the main text can focus on the averaged training trend.

Table 21: **Task-wise post-training benchmark scores.** This table expands the main-text summary in Table 3. Scores are accuracies; ΔAvg is relative to the baseline at the same model scale.

Model	Setting	ARC-E	BoolQ	HSwag	OBQA	PIQA	WinoGr	Avg	ΔAvg
1.4B	Baseline	56.0	65.1	60.5	34.7	75.8	58.7	58.5	0.0
	Attn Para-Rem.	57.1	63.9	61.3	35.3	76.0	59.1	58.8	+0.3
	V-Para Rem.	58.3	62.8	62.1	35.9	76.3	59.8	59.2	+0.7
2.7B	Baseline	58.4	61.2	66.0	37.1	76.5	61.8	60.2	0.0
	Attn Para-Rem.	59.3	62.8	66.7	37.7	77.0	62.1	60.9	+0.7
	V-Para Rem.	60.4	64.4	67.1	38.2	77.6	62.6	61.7	+1.5

Gated parallel-scaling diagnostic. Figure 8 compares 1.4B training traces for baseline training, fixed V-Para Rem., and a gated parallel-scaling variant. The gated variant predicts a tokenwise scalar gate from the hidden state using a learned $d \rightarrow 1$ projection and uses it to modulate the parallel component. This auxiliary run does not improve consistently over fixed removal, so we use it only as a diagnostic variant.

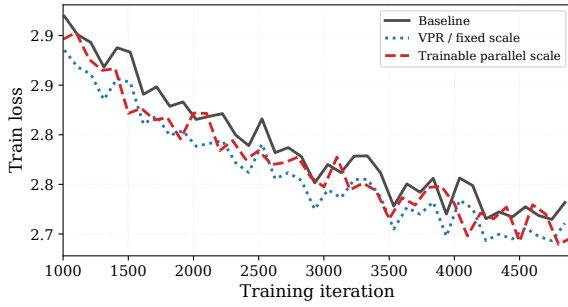


Figure 8: **Ablation study on parallel-scale adjustment during 1.4B pretraining.** Curves compare baseline training, fixed V-Para Rem., and a gated parallel-scale adjustment.

G Extended Compression Geometry

This section adds three diagnostics for the compression analysis in Figure 5. All quantities use the same normalization: the compressed-minus-baseline update error is decomposed relative to the baseline update, and each component magnitude is divided by $\|\Delta_{\text{base}}\|$. Table 22 shows that, for attention-output compression, total local error is strongly aligned with the perpendicular component across layers within each compression setting. The remaining figures extend this error decomposition to block and MLP outputs.

Table 22: **Attention-output total error is mostly aligned with perpendicular error.** For each compression setting, we correlate the layerwise total local error $\|e\|/\|\Delta_{\text{base}}\|$ with its perpendicular and parallel components. Perp./Total and Para./Total report mean component-to-total ratios across layers.

Setting	$r(e, e_{\perp})$	$\rho(e, e_{\perp})$	$\rho(e, e_{\parallel})$	Perp./Total	Para./Total
Quantization	0.998	0.990	0.596	0.980	0.168
Unstructured	0.970	0.909	0.810	0.891	0.439
4:8	0.986	0.986	-0.135	0.887	0.415
2:4	0.991	0.979	-0.432	0.875	0.425

Relation to layer-drop cosine. The parallel/perpendicular view also explains a common layer-drop diagnostic (He et al., 2026). Let x be the incoming representation and let a layer update decompose as $\Delta = \Delta_{\parallel} + \Delta_{\perp}$, with $\Delta_{\parallel} = \alpha x$ and $\|\Delta_{\perp}\| = \beta\|x\|$. Then

$$\begin{aligned} \cos(x, x + \Delta) &= \frac{1 + \alpha}{\sqrt{(1 + \alpha)^2 + \beta^2}} \\ &= \frac{1}{\sqrt{1 + \left(\frac{\beta}{1 + \alpha}\right)^2}}, \end{aligned}$$

assuming $1 + \alpha > 0$. Therefore the usual layer-drop quantity $1 - \cos(x, x + \Delta)$ is a monotone function of $\beta/(1 + \alpha)$, and for small angles is approximately $\frac{1}{2}(\beta/(1 + \alpha))^2$. In this sense, layer-drop cosine summarizes two geometric quantities: the perpendicular update ratio β and the parallel rescaling $1 + \alpha$. It assigns larger relevance to layers whose updates induce a larger relative deviation from the incoming direction. Increasing β directly enlarges this deviation, while $1 + \alpha$ sets the residual scale in the denominator. A larger positive α therefore reduces the effective perpendicular ratio, whereas a negative α increases it. This interpretation is consistent with our view that perpendicular components drive representation evolution, while parallel components modulate the scale against which this directional change is measured. Figure 9 shows the baseline-update magnitude that supports this interpretation.

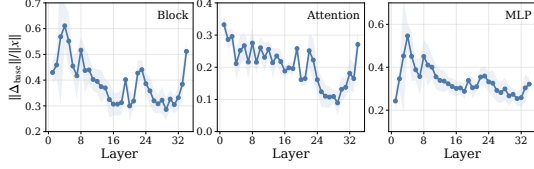


Figure 9: **Baseline layer updates are small relative to the residual stream.** The uncompressed update magnitude $\|\Delta\|/\|x\|$ remains well below one across layers and branches. Since the parallel coefficient α controls the residual rescaling term $1 + \alpha$, layer-drop cosine reflects the effective perpendicular ratio $\beta/(1 + \alpha)$, not β alone.

Overall pruning-error rankings. The compression results above separate error into parallel and perpendicular parts after measuring a local approximation error. *We next check whether standard error-based pruning scores already rank the same coordinates as the perpendicular component.* For each model, we compute the total pruning-error score and its perpendicular component on C4 calibration examples, then compare their rankings across attention residual columns, MLP residual neurons, and value-projection columns. The two rankings are nearly identical: the largest-error entries are also the largest perpendicular-error entries. This helps explain why WANDA-style total-error objectives are already strong baselines for perpendicular damage. We therefore use these measurements as diagnostics for interpreting compression behavior rather than as a separate compression method contribution.

Table 23: **Overall pruning error is nearly aligned with its perpendicular component.** Each row averages over attention residual columns, MLP residual neurons, and value-projection columns when available. Top- k reports the overlap between the largest- k % entries ranked by total error and perpendicular error.

Model	Spearman $\uparrow$	Perp./Total	Para./Total	Top-1%	Top-10%
Gemma-2-2B	1.000	0.999	0.030	0.995	0.998
Llama-3.2-1B	1.000	0.999	0.032	0.997	0.999
Llama-3.2-1B-Inst.	1.000	0.999	0.031	0.996	0.999
Phi-2	1.000	0.999	0.030	0.994	0.998
Qwen3-0.6B-Base	1.000	0.998	0.044	0.989	0.998
Qwen3-1.7B	1.000	0.999	0.038	0.990	0.998
Qwen3-4B	1.000	0.999	0.032	0.996	0.999
Qwen3-8B	1.000	0.999	0.032	0.995	0.999

Table 23 compares candidate sets rather than numerical error values. We evaluate two controls over 23 available model-space groups spanning eight models and attention residual columns, MLP residual neurons, and value-projection columns, using ten random seeds per available group. Phi-2

lacks the MLP-neuron artifact. Uniform candidate sets give the chance-level overlap expected at each selection fraction. A stronger isotropic control samples a norm-matched random direction before decomposing each error.

Table 24: **Controls for total/perpendicular candidate overlap.** Values average the 23 available model-space groups. Observed overlap exceeds all ten uniform-random seeds in every group. The norm-matched isotropic control matches or exceeds the observed overlap in 22/23 and 21/23 groups, showing that high-dimensional concentration substantially explains the result.

Ranking comparison	Top-1%	Top-10%
Observed total vs. perpendicular	99.40%	99.86%
Uniform-random candidates	0.96%	10.00%
Isotropic norm-matched direction	99.94%	99.95%

The observed ranking is therefore highly non-random relative to uniform candidate selection but is not stronger than the norm-matched directional null. We use Table 23 as a descriptive consistency check, not evidence that perpendicular error is uniquely learned or more predictive than total-error magnitude.

We separately test whether the component weighting preserves downstream quality. Table 25 reports matched task averages from the same evaluation manifests. Across the six models, residual-perpendicular weighting changes the downstream average by -0.24 to $+0.38$ points relative to standard WANDA: five models improve, while one decreases by 0.24 points. Thus, the directional metric preserves standard WANDA quality and sometimes yields modest gains, rather than establishing consistent superiority.

Table 25: **Matched downstream quality under component-weighted WANDA.** Scores are unweighted means over OpenBookQA, PIQA, RTE, WinoGrande, BoolQ, ARC-Challenge, HellaSwag, and MMLU. Delta is residual-perpendicular minus standard WANDA. Residual-parallel is reported for the five models with matched all-linear coverage. Small positive deltas are descriptive and are not claimed as universal improvements.

Model	WANDA	Residual-perp.	Delta	Residual-para.
Gemma-2-2B	55.73	55.77	+0.04	54.72
Llama-3.2-1B	43.27	43.43	+0.16	42.67
Llama-3.2-1B-Inst.	45.38	45.14	-0.24	44.78
Phi-2	61.44	61.48	+0.04	—
Qwen3-1.7B	54.17	54.55	+0.38	51.76
Qwen3-4B	61.45	61.56	+0.12	60.66

On Qwen3-1.7B/4B, residual-perpendicular weighting changes only 0.0098%/0.0081% of pruned weights relative to standard WANDA, while residual-parallel weighting changes 8.81%/9.68%. Across the five models with matched residual-parallel evaluations, that variant underperforms the perpendicular variant by 0.35–2.79 points. These comparisons support the narrow conclusion that perpendicular-aware weighting is performance-preserving in the tested settings, whereas parallel error is less discriminative. They do not establish that perpendicular error universally outperforms total error.

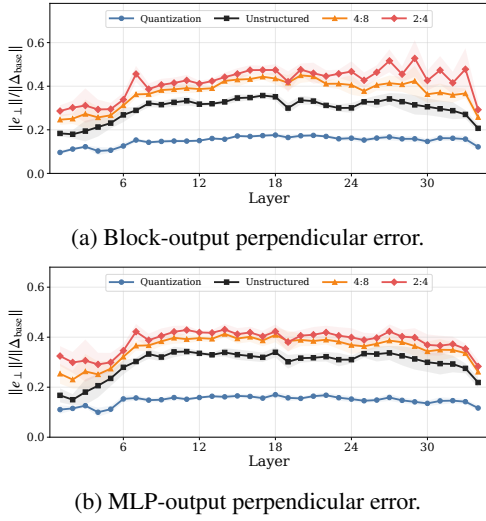


Figure 10: **Compression-geometry breakdown across branches.** Top/bottom: normalized perpendicular component $\|e_{\perp}\|/\|\Delta_{\text{base}}\|$ of block- and MLP-output compression error.

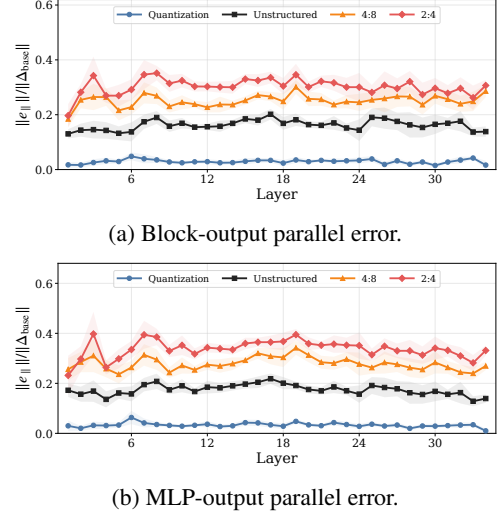


Figure 11: **Parallel-error breakdown under compression.** Top/bottom: normalized parallel component $\|e_{\parallel}\|/\|\Delta_{\text{base}}\|$ of block- and MLP-output compression error. The attention-output parallel view appears alongside the attention-output perpendicular view in Figure 5.